



Chrono

Real-time polyglot database pipeline connecting stock markets, forex, commodities, and macroeconomic intelligence into a spatial 3D command dashboard.



The Problem: Fragmented Financial Data

Scattered Sources

Stock tickers, FX rates, and commodity prices siloed across separate tabs and disconnected systems — no unified view of interconnected markets.

Delayed Context

Hours of latency between price movements and explanatory market intelligence, leaving analysts reacting rather than anticipating.

No Regional Perspective

Lack of cross-border macroeconomic visibility and unified global market awareness prevents holistic decision-making.

The Solution: Single Unified Financial Radar



1-Stop Control Room

Centralized 3D interactive globe with real-time market data, news feeds, and economic indicators in one immersive interface.



Plain-English AI Market Summaries

ML-powered sentiment classification and urgency scoring on breaking news — Bullish, Bearish, or Neutral in real time.



Interactive 3D Global Hub

Raycasting-enabled country navigation with localized financial control panels triggered by mouse intersection detection.



Hybrid Data Streams

REST/GraphQL for historical data; persistent WebSocket connections for real-time tick updates and breaking news.

Key Value & Efficiency Metrics

<50ms

Ticker Latency

Sub-50 millisecond price tick delivery to the 3D canvas via Redis Pub/Sub and WebSocket streaming.

100+

Global Instruments

Real-time coverage across stocks, FX pairs, commodities, and macroeconomic indicators worldwide.

1-Click

Economic Assessment

Instant country-level financial snapshot with integrated news context via raycasting navigation.

80%

Analyst Time Saved

Automated data aggregation and sentiment classification eliminates manual research overhead.

Layer 1: User Access & Presentation Layer

ARCHITECTURE LAYER 1 OF 3



SPA Authentication

JWT / HTTP-only session cookies; credentials verified against **PostgreSQL (DB1)** with client-side route guards.

3D Spatial Canvas

Full-screen WebGL globe via **Globe.gl + Three.js**; GeoJSON country boundaries and 3D instanced mesh markers for financial hubs.

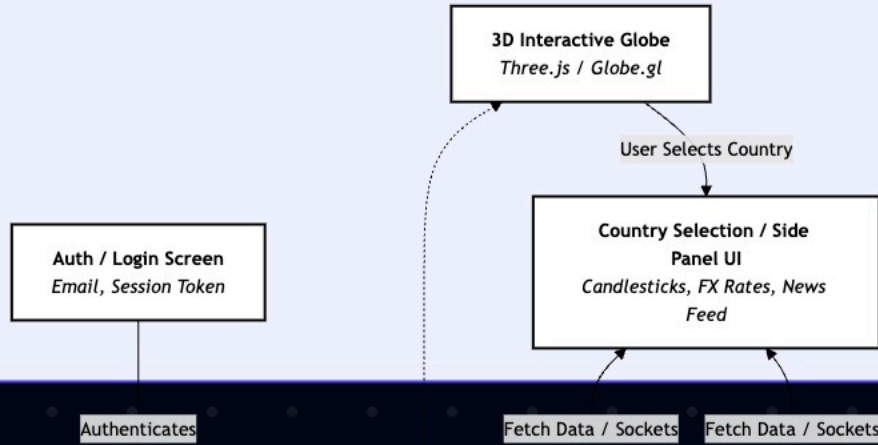
Raycasting Navigation

Mouse intersection resolves ISO country codes; smooth camera interpolation triggers localized financial control panels.

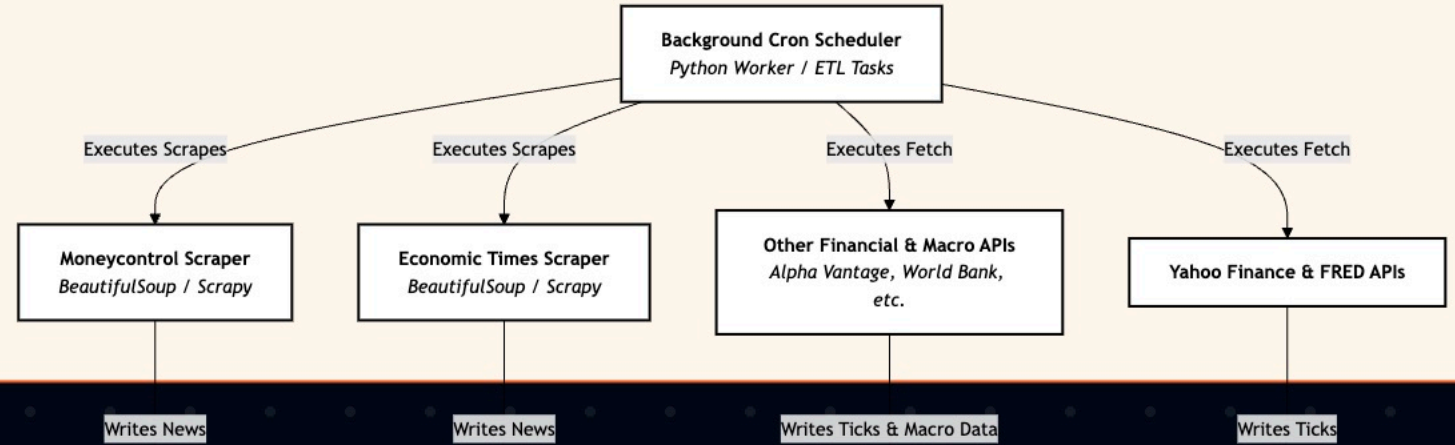
Hybrid Data Fetching

REST/GraphQL → TimescaleDB historical series;
WebSocket → Redis Pub/Sub for live ticks and sentiment.

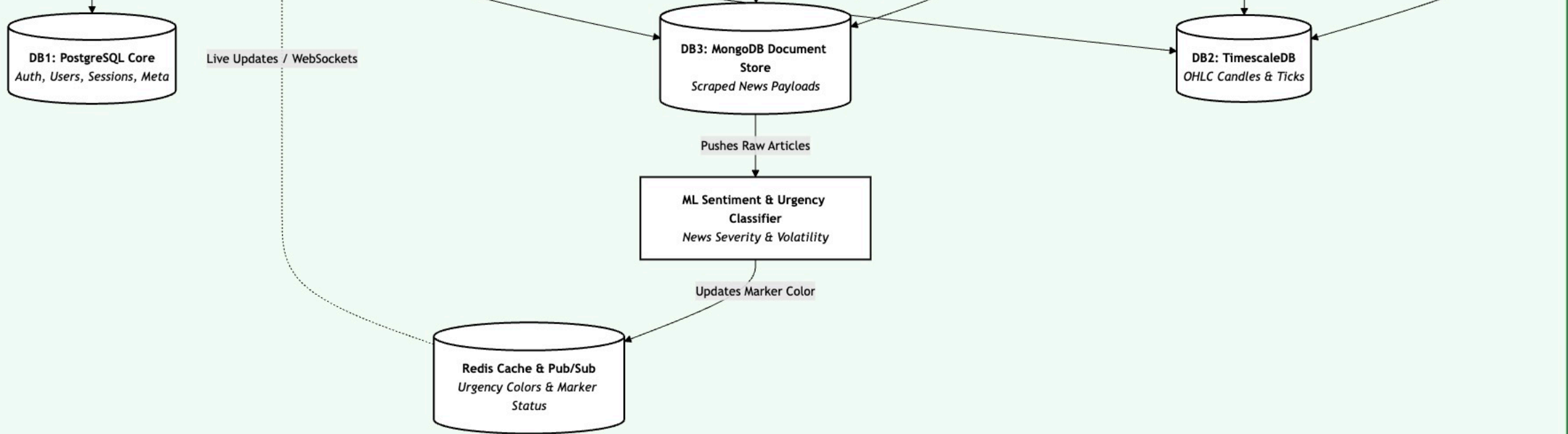
LAYER 1: User Access & Presentation Layer



LAYER 2: Data Processing & Ingestion Pipeline

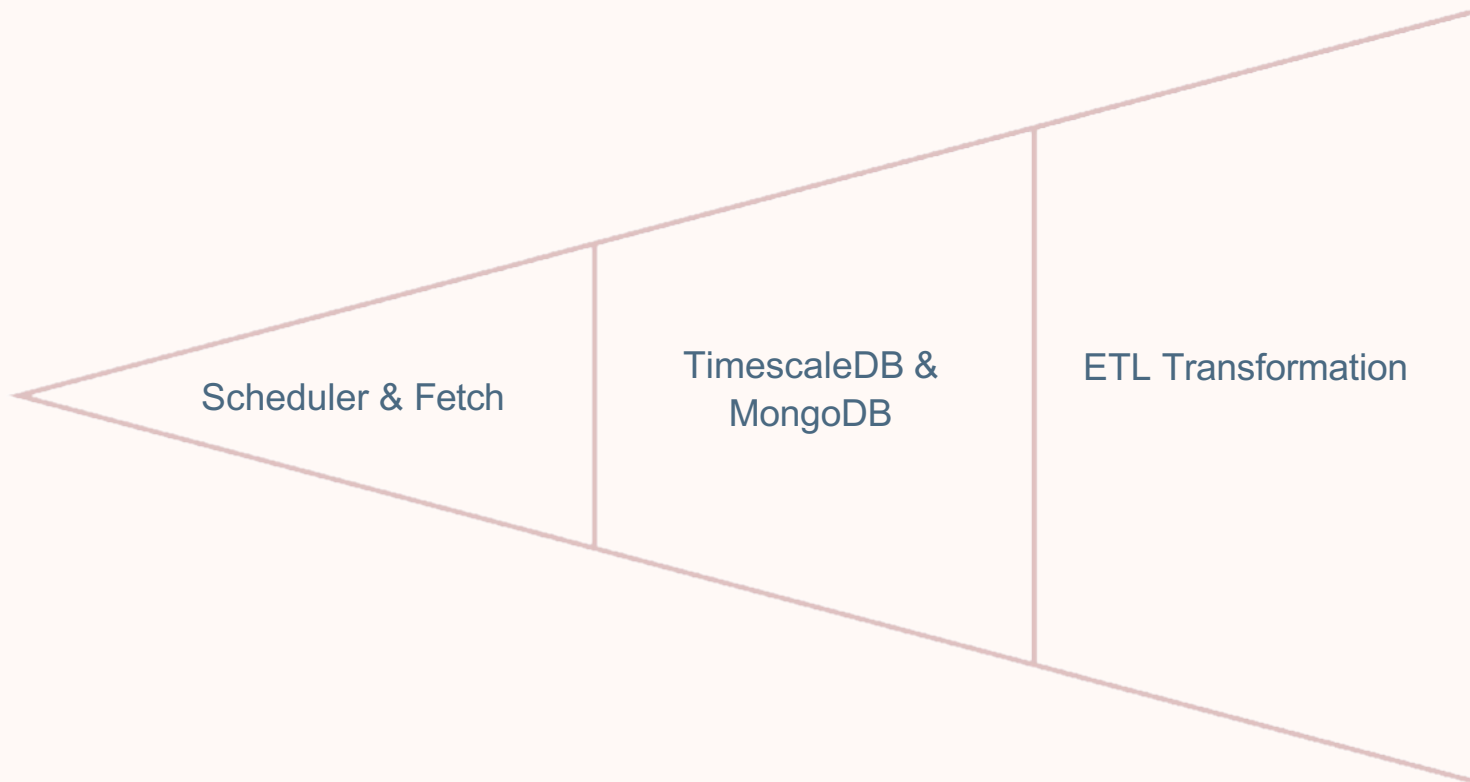


LAYER 3: Polyglot Persistence Engine



Layer 2: Data Processing & Ingestion Pipeline

ARCHITECTURE LAYER 2 OF 3



Python orchestration via **APScheduler** / **Celery** / **asyncio** decouples external fetching from the frontend. Structured connectors poll Yahoo Finance, FRED, Alpha Vantage, and World Bank APIs for OHLC candlesticks and FX rates. Unstructured scrapers (BeautifulSoup, Scrapy, Playwright) crawl Reuters, Economic Times, and Moneycontrol for breaking headlines. The ETL pipeline normalizes JSON/HTML, converts timestamps to UTC, tags news with ISO codes, then routes [time-series ticks](#) → [TimescaleDB \(DB2\)](#) and [unstructured news](#) → [MongoDB \(DB3\)](#).

Layer 3: Polyglot Persistence Engine

ARCHITECTURE LAYER 3 OF 3

PostgreSQL (DB1)

ACID-compliant relational authority — user accounts, hashed credentials, session tokens, subscription tiers, ISO codes, coordinates, central bank profiles.



TimescaleDB (DB2)

Partitioned hypertables for OHLC candlesticks, intraday ticks, FX rates; continuous aggregates auto-calculate moving averages and volatility metrics.



MongoDB (DB3)

Schema-less BSON document store for scraped news, press releases, and raw API payloads; full-text search and geo/spatial queries by country code.



Redis Cache & Pub/Sub

Sub-millisecond caching of country urgency flags; Pub/Sub broker pushes ML classification updates to the 3D globe via WebSocket channels.



Database Deep Dive: PostgreSQL vs TimescaleDB

PostgreSQL — DB1: Relational Core

Manages authentication, user profiles, session states, and master reference data.

- Full **ACID compliance** and strict foreign key constraints prevent orphaned records and data drift
- Authentication and user state cannot tolerate eventual consistency — relational integrity is non-negotiable
- Shared driver ecosystem bridges seamlessly into TimescaleDB

TimescaleDB — DB2: Time-Series Engine

Automatic time-based partitioning via **hypertables** keeps index sizes small and sustains high-speed writes into millions of rows.

- Native SQL **continuous aggregates** calculate 50-day moving averages and hourly volume metrics in the background
- Standard PostgreSQL hits performance walls at high-frequency tick volume — TimescaleDB solves this while maintaining full SQL compatibility
- Zero new driver overhead: identical connection string to DB1

Database Deep Dive: MongoDB & Redis

1

MongoDB — DB3: Document Store

Schema-less **BSON** repository for unstructured news articles, press releases, and raw API payloads. Web scrapers and external APIs constantly change format — MongoDB handles polymorphic JSON natively without rigid schema migrations.

Built-in **full-text search** and geo/spatial query support enable fast keyword filtering by ISO country code and publication timestamp before ML processing.

2

Redis — Cache & Pub/Sub Broker

Sub-millisecond read latency for country urgency status flags eliminates repetitive database reads. Native Pub/Sub engine intercepts ML classification tags and instantly pushes color-code updates to the 3D globe via WebSockets.

Why Redis over RabbitMQ/Kafka?

Redis provides both key-value caching AND message brokering in a single container — Kafka/RabbitMQ lack caching. Shared Redis instance ensures consistent state across all backend workers.



Database Systems Architectural Comparison Matrix

Storage Engine	Data Type	Primary Role	Core Reason for Choice
PostgreSQL (DB1)	Structured / Relational	Auth, Users, Sessions, Master Metadata	Strict ACID compliance, relational integrity, shared ecosystem with TimescaleDB
TimescaleDB (DB2)	Structured Time-Series	High-frequency OHLC ticks, Candles, FX rates	Automatic time-based partitioning (hypertables), native SQL continuous aggregates
MongoDB (DB3)	Unstructured Documents	Scraped news stories, raw API payloads	Flexible schema-less BSON structure, fast text indexing, polymorphic JSON
Redis	In-Memory Key-Value / Streams	Urgency status cache, WebSocket Pub/Sub	Sub-millisecond latency, built-in Pub/Sub broker for live UI updates

❏ Each database is chosen for its access pattern — no single engine could optimally serve all four roles simultaneously. This is the polyglot rationale.

Literature Survey

1. Halili et al. (2025)

Polyglot Persistence in Microservices

Compares relational, document, key-value, column-family, and graph databases across scalability, consistency, and operational complexity.

Chrono Relevance: Industry evidence from Netflix, Uber, and Shopify validates combining PostgreSQL, TimescaleDB, MongoDB, and Redis instead of relying on a single database.

2. Van Landuyt et al. (2023)

Performance of Polyglot Persistence

Benchmarks dedicated polyglot persistence against multi-model NoSQL databases to evaluate performance and workload specialisation.

Chrono Relevance: Supports separating relational, time-series, and document workloads rather than adopting a single multi-model database.

3. Gu et al. (2024)

FinBERT-LSTM Stock Prediction

Integrates FinBERT-based financial news sentiment with LSTM models for improved stock price forecasting.

Chrono Relevance: Provides the foundation for Chrono's AI Market Summaries, enabling real-time Bullish / Bearish / Neutral sentiment classification from financial news.

Conclusion & Technical Roadmap

A three-layer polyglot persistence system unifies fragmented financial data through databases optimized for their specific access patterns and data characteristics.

